

SHASHWAT YADAV

+91 6393355490 • shashwatyadav101@gmail.com • [LinkedIn](#) • [GitHub](#) • Bengaluru, India

PROFILE

Applying for Amazon ML Summer School 2026 (MLSS 2026) — an AI/ML engineering student at a recognized institute (2027 BE, CGPA 9.15/10) with production internship experience building LLM-driven applications, RAG pipelines, and ML-integrated data systems. IEEE-published researcher, two-time hackathon winner, and hands-on practitioner of supervised learning, deep learning, NLP, vector search, and cloud computing (AWS). Strong mathematical foundation in probability, linear algebra, and statistics — directly aligned with Amazon ML Summer School modules: supervised learning, deep neural networks, probabilistic graphical models, dimensionality reduction, sequential models, and generative AI.

EDUCATION

B.E. — Artificial Intelligence & Machine Learning

Aug 2023 – May 2027

Dayananda Sagar Academy of Technology & Management, Bengaluru

CGPA: 9.15 / 10

Relevant Coursework: Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, DBMS, Computer Networks, Operating Systems, System Design

12th Standard — Physics, Chemistry, Mathematics (CBSE)

Kendriya Vidyalaya

10th Standard (CBSE)

Kendriya Vidyalaya

EXPERIENCE

AI Engineer Intern — Backend & ML Systems

Jan 2026 – May 2026

Hasprana Health Care Solutions · Bengaluru, India

- Designed and deployed ML models for predictive analytics in production, improving forecasting accuracy by 15% and operational efficiency by 20% — hands-on supervised learning in an enterprise setting.
- Built Python & Java automation pipelines for healthcare data processing, reducing manual effort by 30% and enabling real-time analytics across 10,000+ daily records.
- Constructed high-throughput backend services handling 5+ concurrent data streams with HIPAA-compliant privacy controls and strict fault-tolerance.
- Integrated LLM modules into enterprise applications within a 6–8 person Agile/Scrum squad, consistently delivering 100% of sprint goals across a 5-month engagement.
- Authored 40+ unit and integration tests achieving 95%+ code coverage on critical data-pipeline modules.

ML & AI PROJECTS

Medical RAG System [LINK](#)

Python · LangChain · ChromaDB · Llama · Groq API

- Developed a production-ready Retrieval-Augmented Generation pipeline grounded in clinical guidelines — end-to-end ingestion, vectorization, semantic retrieval, and LLM inference with near-zero hallucination rate across 500+ test queries.
- Employed advanced text chunking, embedding strategies, and prompt engineering, achieving ~40% latency reduction over naive baseline — directly applicable to generative AI & LLMs module topics.

Medical Report Simplifier — 1st Place, BMS Hackathon [LINK](#)

Django · LLM Inference · Python · PDF

Generation

- developed and shipped a full-stack AI-powered web application automating patient PDF report generation using LLM inference, reducing report-reading time by 50% — outperformed all 20+ competing teams.
- Architected end-to-end pipeline: user request handling → LLM inference → structured PDF output, showcasing end-to-end deep learning integration under production constraints.

NexusSearch — Sharded Vector Search Engine [LINK](#)

Python · gRPC · Horizontal Sharding · ANN

- Architected a custom vector search engine with approximate nearest neighbor (ANN) indexing and horizontal sharding, cutting average query latency by 35% over single-node baseline — practical application of dimensionality reduction and embedding theory.
- Defined gRPC API contracts across 4 independent microservices with fault tolerance across nodes; achieved sub-50ms latency on 100K+ record retrieval.

Sandboxed Algorithmic Judge & AI Code Reviewer [LINK](#) Python · Groq/Llama · LLM · Docker · REST API

- ▶ Designed an LLM-driven automated code review pipeline analyzing algorithmic correctness, time/space complexity, and code quality across 5+ languages — unsupervised and generative AI leveraged for structured analysis.
- ▶ Created a secure sandboxed execution environment evaluating 50+ test cases per submission in under 2 seconds with zero cross-submission data leakage.

AeroCache — Clustered In-Memory Caching System [LINK](#) Java · Consistent Hashing · LRU/LFU Eviction · Multi-Node Systems

- ▶ Built a fault-tolerant multi-node caching system with consistent hashing for zero-downtime node scaling, sustaining 99.9% uptime under load testing across 10 independent nodes.

TECHNICAL SKILLS

ML / AI: Supervised Learning, Deep Learning, NLP, RAG Pipelines, LangChain, LLMs, Prompt Engineering, Semantic Search, ChromaDB, Llama, Groq API, scikit-learn

Math Foundations: Linear Algebra, Probability & Statistics, Matrix Operations, Hypothesis Testing, Vector Spaces, Bayesian Inference

Languages: Python (primary), Java, SQL, JavaScript, C

Frameworks: Django, FastAPI, REST APIs, gRPC, Microservices, Node.js

Databases: PostgreSQL, MongoDB, ChromaDB (Vector DB), MySQL, LRU/LFU Caching

Cloud & DevOps: AWS Cloud Computing (EC2, Lambda, S3, API Gateway), Docker, Git/GitHub, Linux/CLI, Agile/Scrum, Unit Testing (pytest)

Certifications: Machine Learning Fundamentals (Infosys Springboard) · Introduction to NLP (Infosys Springboard) · Certified System Administrator (ServiceNow)

ACHIEVEMENTS & PUBLICATIONS

- ▶ BMS Hackathon — 1st Place: Medical Report Simplifier (LLM + Django), outperformed all 20+ competing teams; recognised for practical impact and technical depth.
- ▶ Smart India Hackathon 2025 — Institutional Nominee: Selected from 100+ teams for innovative real-world AI/ML technical solution.
- ▶ IEEE Publication — NQComp 2026: "A Review on AI-Enabled Wildlife Preservation and Management System" — accepted at IEEE International Conference on Next-Gen Quantum and Advanced Computing.